

# Amulet Motion Controller

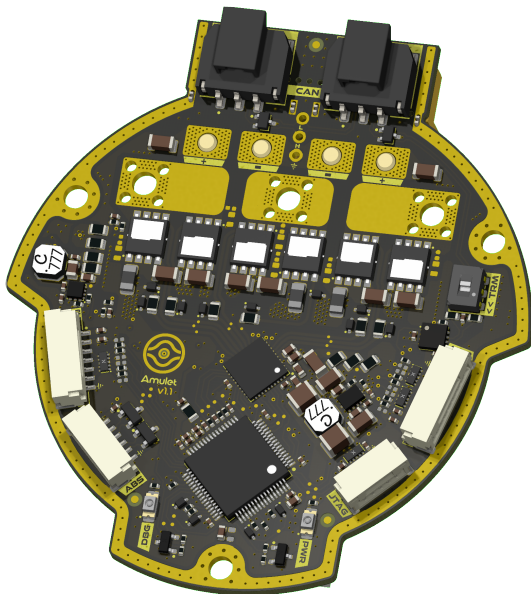
Variant: NO\_SNUBBER

2024-11-19

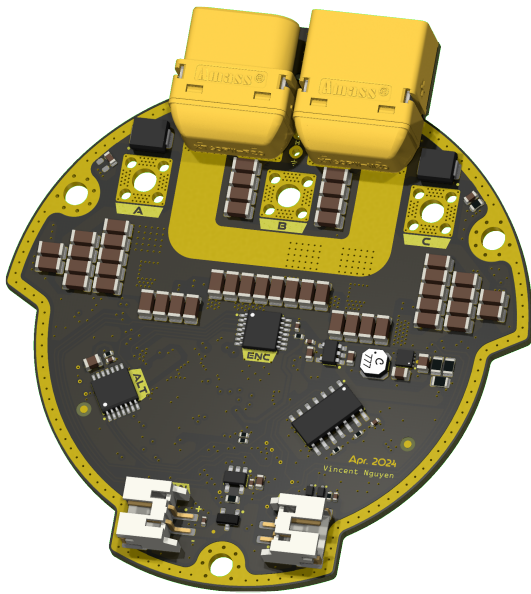
Rev 1.1

| Page | Index                     | Page | Index                     | Page | Index            | Page | Index |
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## TOP VIEW



## BOTTOM VIEW



## DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

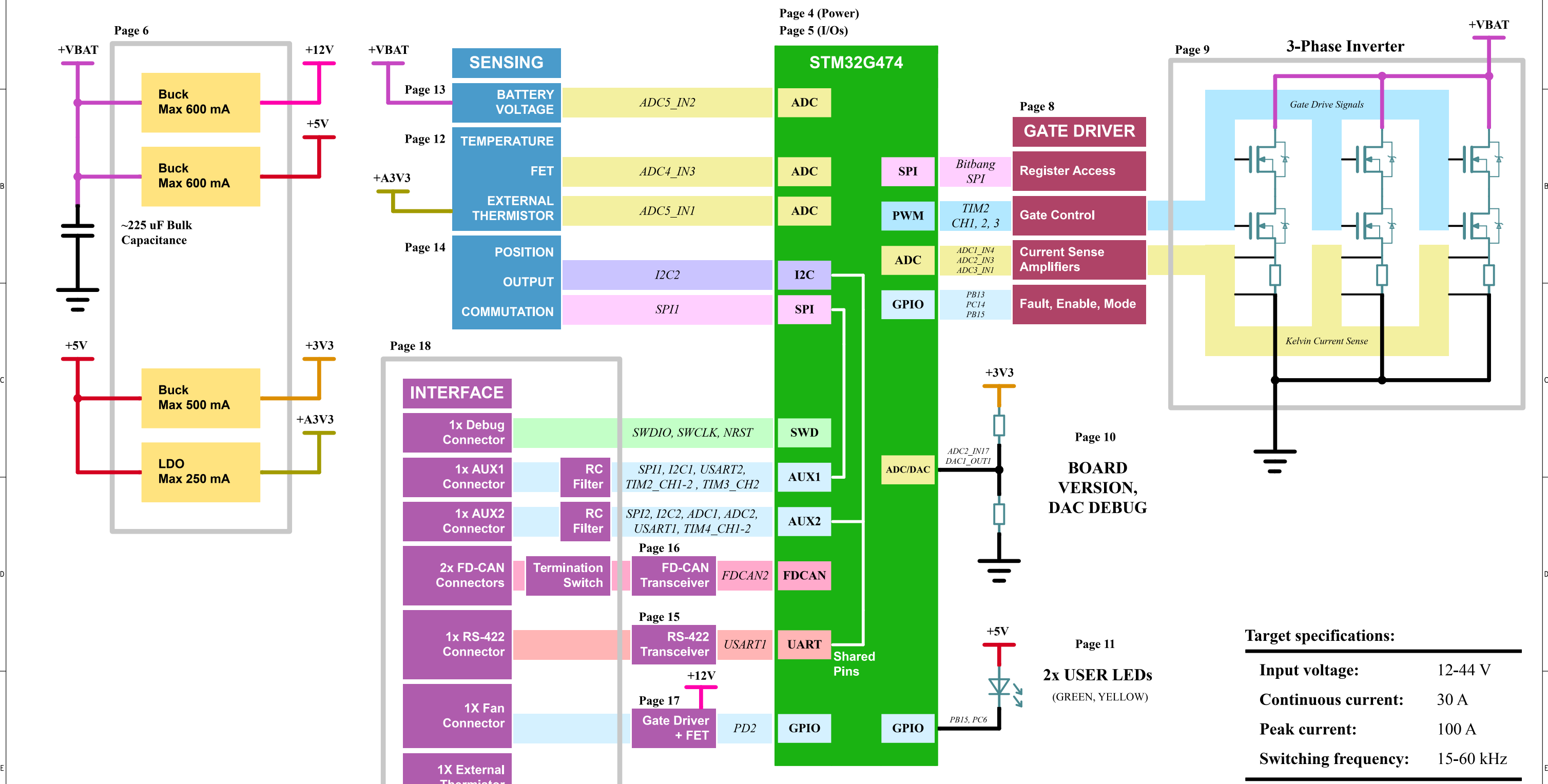
## NOTES

- Schematic based off Josh Pieper's moteus controllers.
- Not fitted components are marked as **X**
- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Date: 13-APR-2024

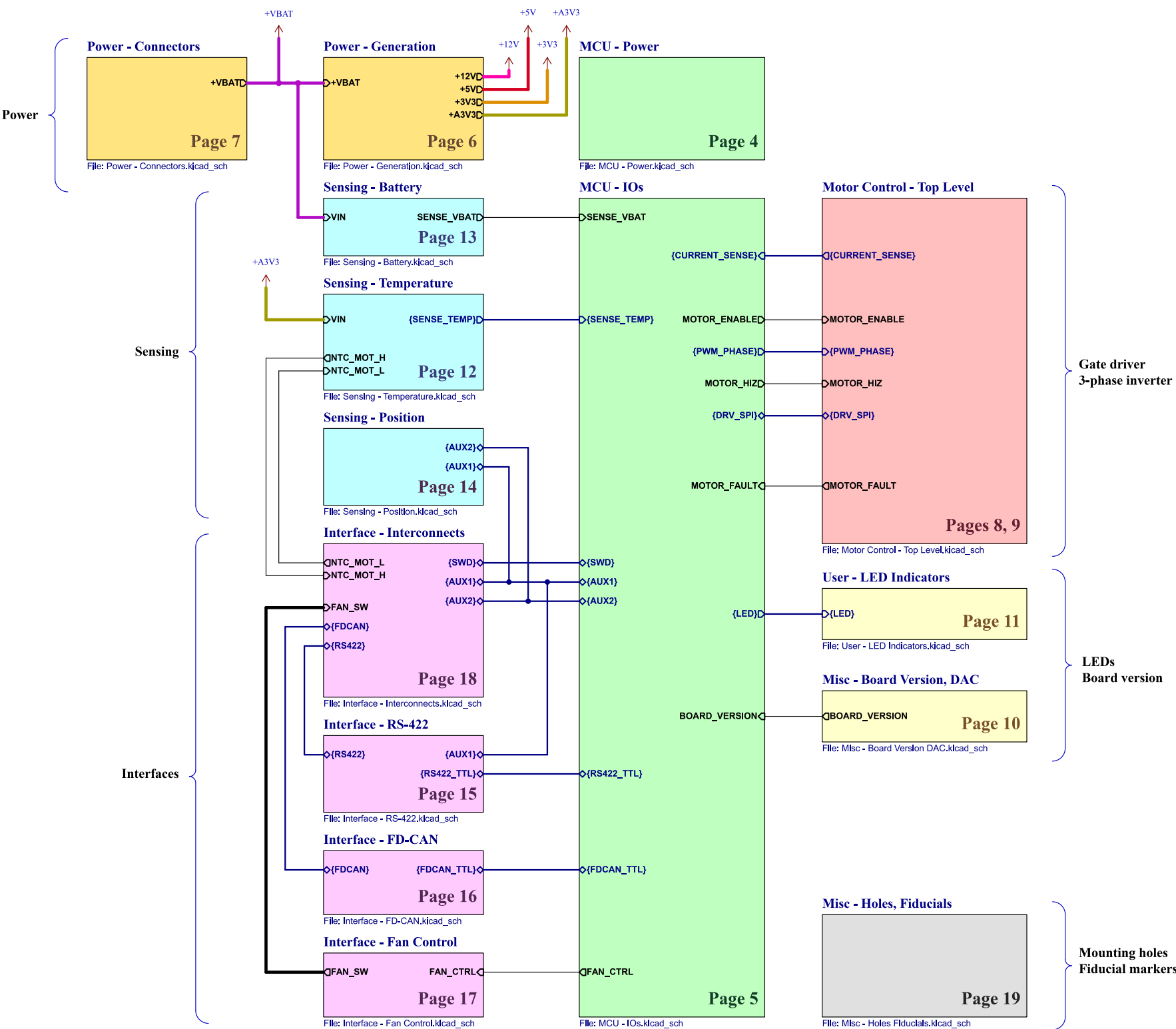
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|  |                            | Board Name:<br>Amulet Motion Controller   |                             | Project Name:<br>Chienpanzé |                   |
|  | Sheet Title:<br>Cover Page | File Name:<br>amulet_controller.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1  |
|  | Sheet Path:<br>/           | Reviewer:                                 |                             | Size:<br>A3                 | Sheet:<br>1 of 21 |

[2] Block Diagram



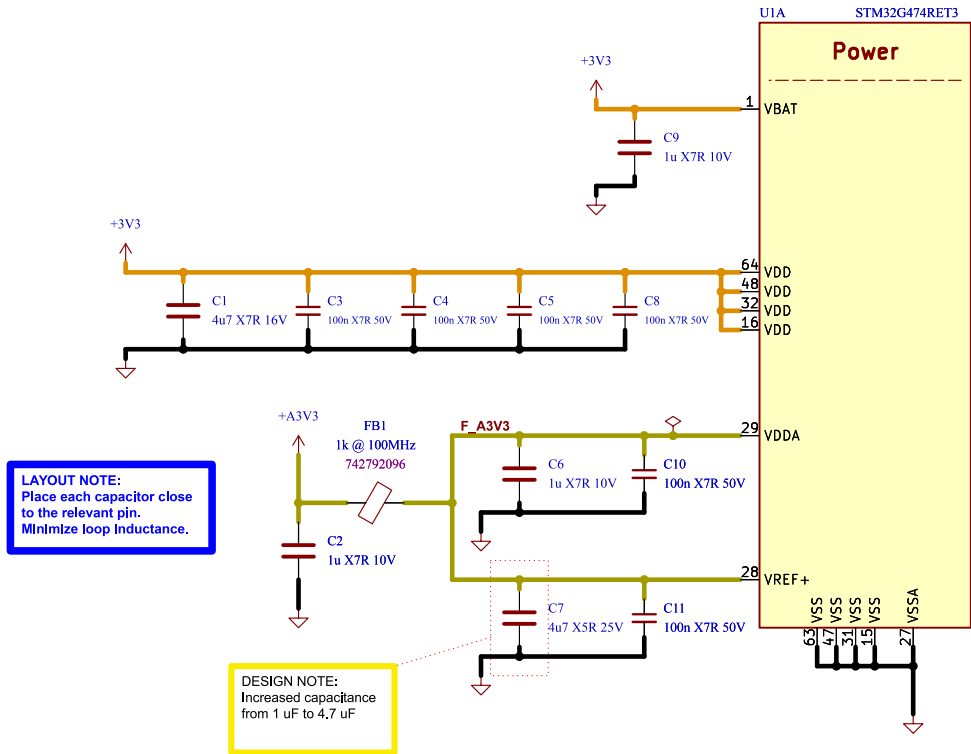
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|  | Comments:                      | Company:<br>EPFL Xplore Research ABC    |                             |  |                   | Variant:<br>NO_SNUBBER |  |
|  |                                | Board Name:<br>Amulet Motion Controller |                             | Project Name:<br>Chienpanzé                                                           |                   |                        |  |
|  | Sheet Title:<br>Block Diagram  | File Name:<br>Block Diagram.kicad_sch   | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13                                                                   | Revision:<br>1.1  |                        |  |
|  | Sheet Path:<br>/Block Diagram/ |                                         | Reviewer:                   | Size:<br>A3                                                                           | Sheet:<br>2 of 21 |                        |  |
|  |                                |                                         |                             |                                                                                       |                   |                        |  |

# [3] Project Architecture



|                                       |                                              |                             |                                                                                       |                             |                        |  |
|---------------------------------------|----------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|------------------------|--|
| Comments:                             | Company:<br>EPFL Xplore Research ABC         |                             |  |                             | Variant:<br>NO_SNUBBER |  |
|                                       | Board Name:<br>Amulet Motion Controller      |                             |                                                                                       | Project Name:<br>Chienpanzé |                        |  |
| Sheet Title:<br>Project Architecture  | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-22                                                                   | Revision:<br>1.1            |                        |  |
| Sheet Path:<br>/Project Architecture/ |                                              | Reviewer:                   | Size:<br>A3                                                                           | Sheet:<br>3 of 21           |                        |  |

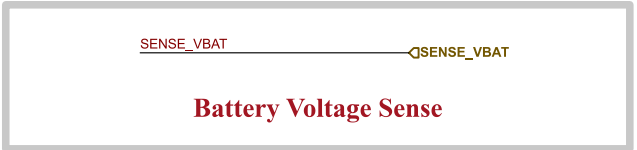
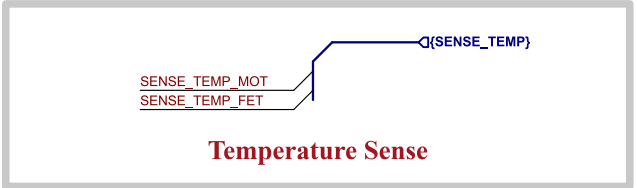
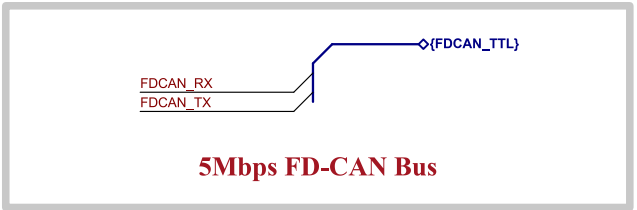
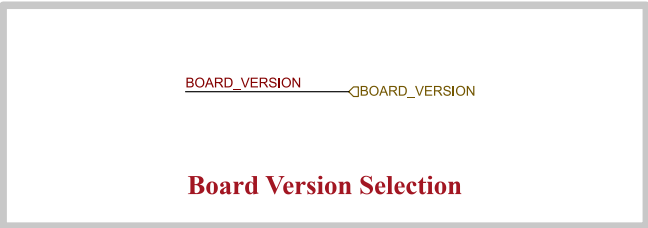
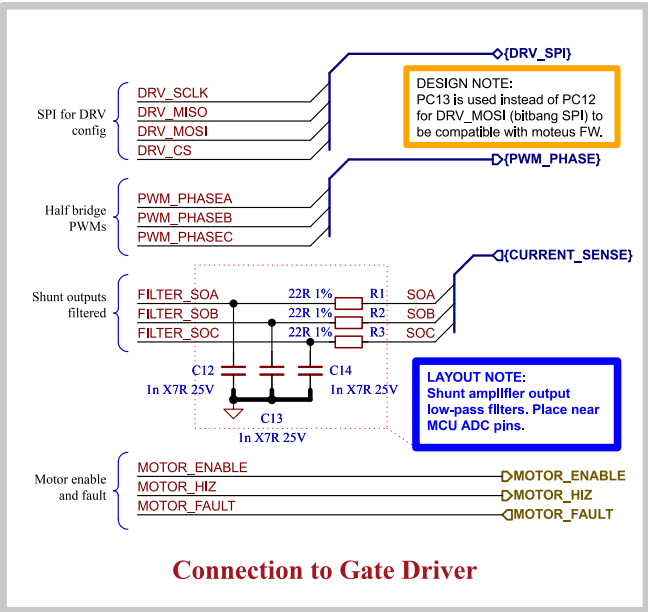
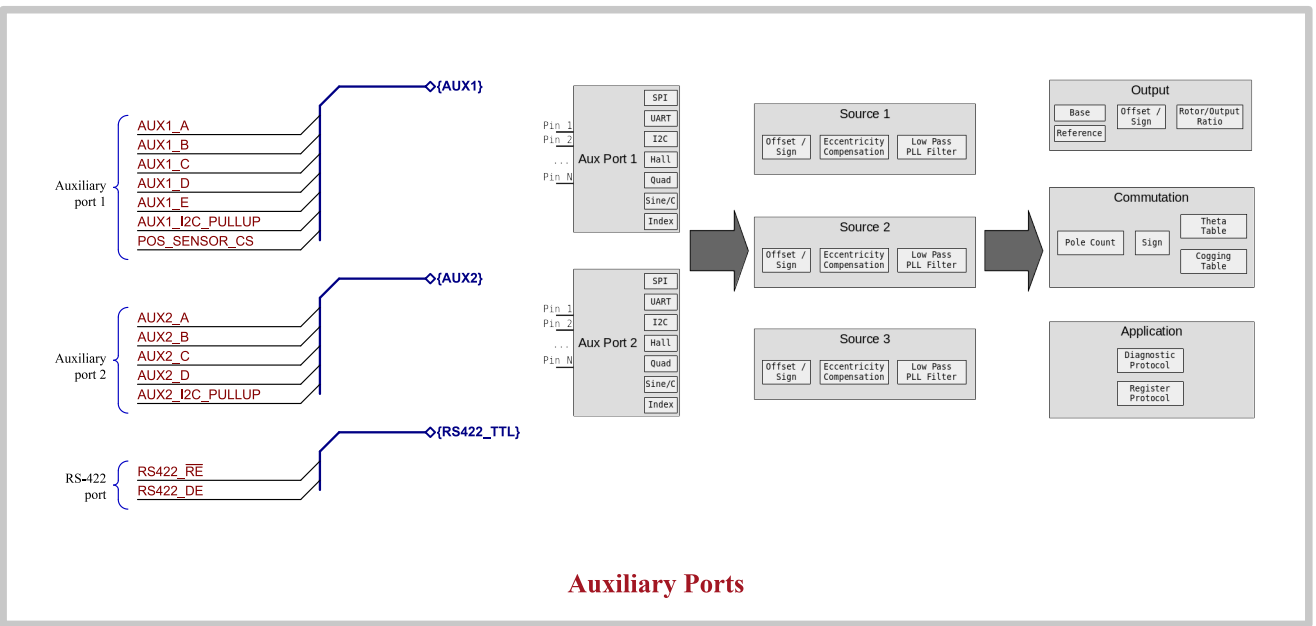
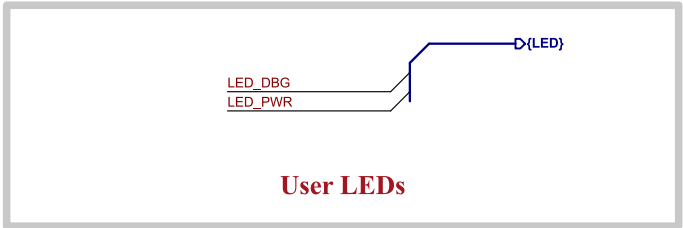
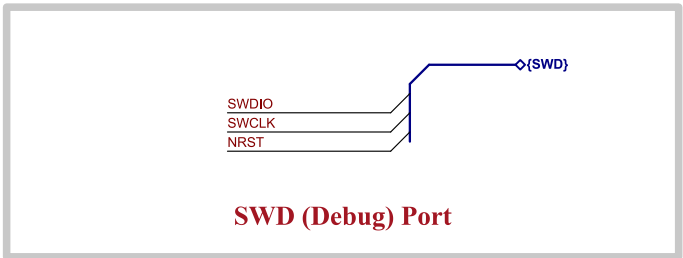
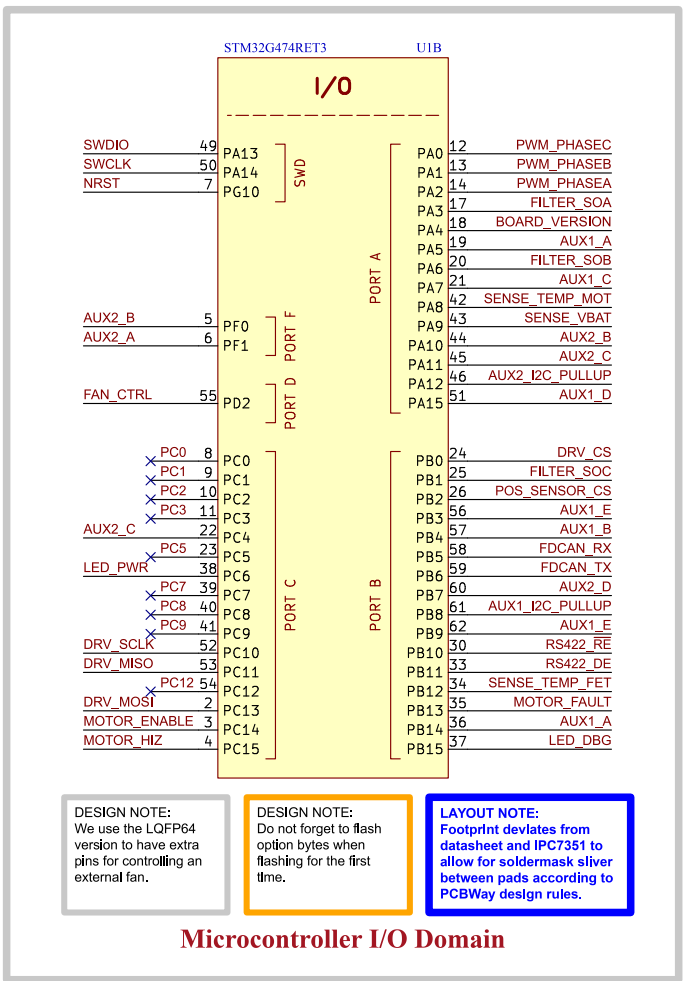
# [4] MCU - Power



Microcontroller Power Supply Scheme

|  |                                                                                                                   |                                                        |                                                                                                                                                          |                                        |                                                |  |
|--|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------------|--|
|  | <div>Comments:</div> <div>AN5346</div> <div>STM32G474 Datasheet p.81</div> <div>J. Pieper ADC investigation</div> |                                                        | <div>Company:</div> <div>EPFL Xplore Research ABC</div> <div></div> |                                        | <div>Variant:</div> <div>NO_SNUBBER</div>      |  |
|  |                                                                                                                   |                                                        | <div>Board Name:</div> <div>Amulet Motion Controller</div>                                                                                               |                                        | <div>Project Name:</div> <div>Chienpanzé</div> |  |
|  | <div>Sheet Title:</div> <div>MCU - Power</div>                                                                    | <div>File Name:</div> <div>MCU - Power.kicad_sch</div> | <div>Designer:</div> <div>Vincent Nguyen</div>                                                                                                           | <div>Date:</div> <div>2023-12-18</div> | <div>Revision:</div> <div>1.1</div>            |  |
|  | <div>Sheet Path:</div> <div>/Project Architecture/MCU - Power/</div>                                              |                                                        | <div>Reviewer:</div>                                                                                                                                     | <div>Size:</div> <div>A4</div>         | <div>Sheet:</div> <div>4 of 21</div>           |  |

# [5] MCU - I/Os



## Gate Driver

|              |      |              |
|--------------|------|--------------|
| MOTOR_ENABLE | TP1  | MOTOR_ENABLE |
| MOTOR_HI_Z   | TP2  | MOTOR_HI_Z   |
| MOTOR_FAULT  | TP3  | MOTOR_FAULT  |
| DRV_SCLK     | TP4  | DRV_SCLK     |
| DRV_MISO     | TP5  | DRV_MISO     |
| DRV_MOSI     | TP6  | DRV_MOSI     |
| DRV_CS       | TP7  | DRV_CS       |
| PWM_PHASEA   | TP8  | PWM_PHASEA   |
| PWM_PHASEB   | TP9  | PWM_PHASEB   |
| PWM_PHASEC   | TP10 | PWM_PHASEC   |
| SOA          | TP11 | SOA          |
| SOB          | TP12 | SOB          |
| SOC          | TP13 | SOC          |

## Debug

|       |      |       |
|-------|------|-------|
| SWDIO | TP14 | SWDIO |
| SWCLK | TP15 | SWCLK |
| NRST  | TP16 | NRST  |

## Auxiliary pins 1

|                 |      |                 |
|-----------------|------|-----------------|
| AUX1_A          | TP17 | AUX1_A          |
| AUX1_B          | TP18 | AUX1_B          |
| AUX1_C          | TP19 | AUX1_C          |
| AUX1_D          | TP20 | AUX1_D          |
| AUX1_E          | TP21 | AUX1_E          |
| AUX1_I2C_PULLUP | TP22 | AUX1_I2C_PULLUP |
| POS_SENSOR_CS   | TP23 | POS_SENSOR_CS   |

## Auxiliary pins 2

|                 |      |                 |
|-----------------|------|-----------------|
| AUX2_A          | TP24 | AUX2_A          |
| AUX2_B          | TP25 | AUX2_B          |
| AUX2_C          | TP26 | AUX2_C          |
| AUX2_D          | TP27 | AUX2_D          |
| AUX2_I2C_PULLUP | TP28 | AUX2_I2C_PULLUP |

## RS-422

|          |      |          |
|----------|------|----------|
| RS422_RE | TP29 | RS422_RE |
| RS422_DE | TP30 | RS422_DE |

## LEDs

|         |      |         |
|---------|------|---------|
| LED_DBG | TP31 | LED_DBG |
| LED_PWR | TP32 | LED_PWR |

## FD-CAN

|          |      |           |
|----------|------|-----------|
| FDCAN_RX | TP33 | FD_CAN_RX |
| FDCAN_TX | TP34 | FDCAN_TX  |

## Fan

|          |      |          |
|----------|------|----------|
| FAN_CTRL | TP35 | FAN_CTRL |
|----------|------|----------|

## Sense

|                |      |                |
|----------------|------|----------------|
| SENSE_TEMP_MOT | TP36 | SENSE_TEMP_MOT |
| SENSE_TEMP_FET | TP37 | SENSE_TEMP_FET |
| SENSE_VBAT     | TP38 | SENSE_VBAT     |

## Test Points

### Comments:

#### References:

Flexible I/O worked examples  
Flexible I/O source configuration

### Sheet Title:

MCU - I/Os

### Sheet Path:

/Project Architecture/MCU - I/Os/

### Company:

EPFL Xplore Research ABC

### Board Name:

Amulet Motion Controller

### File Name:

MCU - I/Os.kicad\_sch

### Designer:

Vincent Nguyen

### Reviewer:

### Variant:

NO\_SNUBBER

### Project Name:

Chienpanzé

### Date:

2023-12-20

### Size:

A3

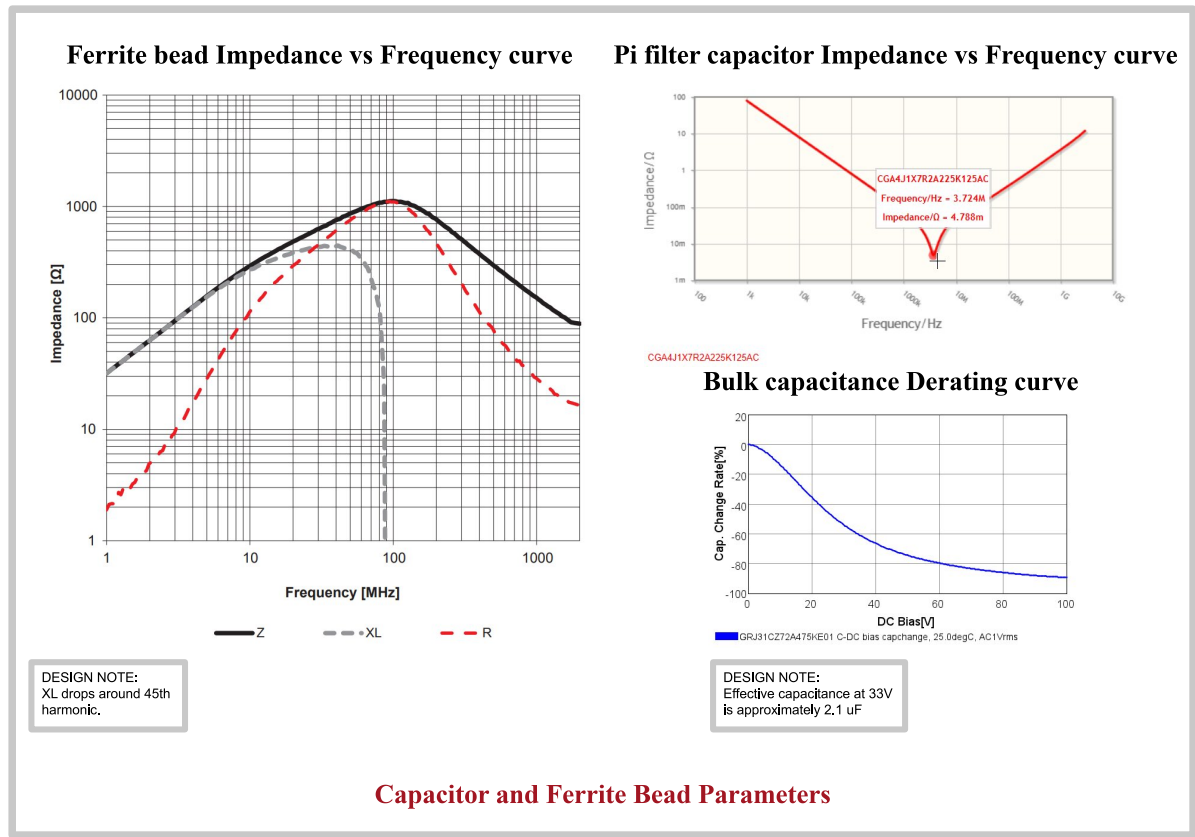
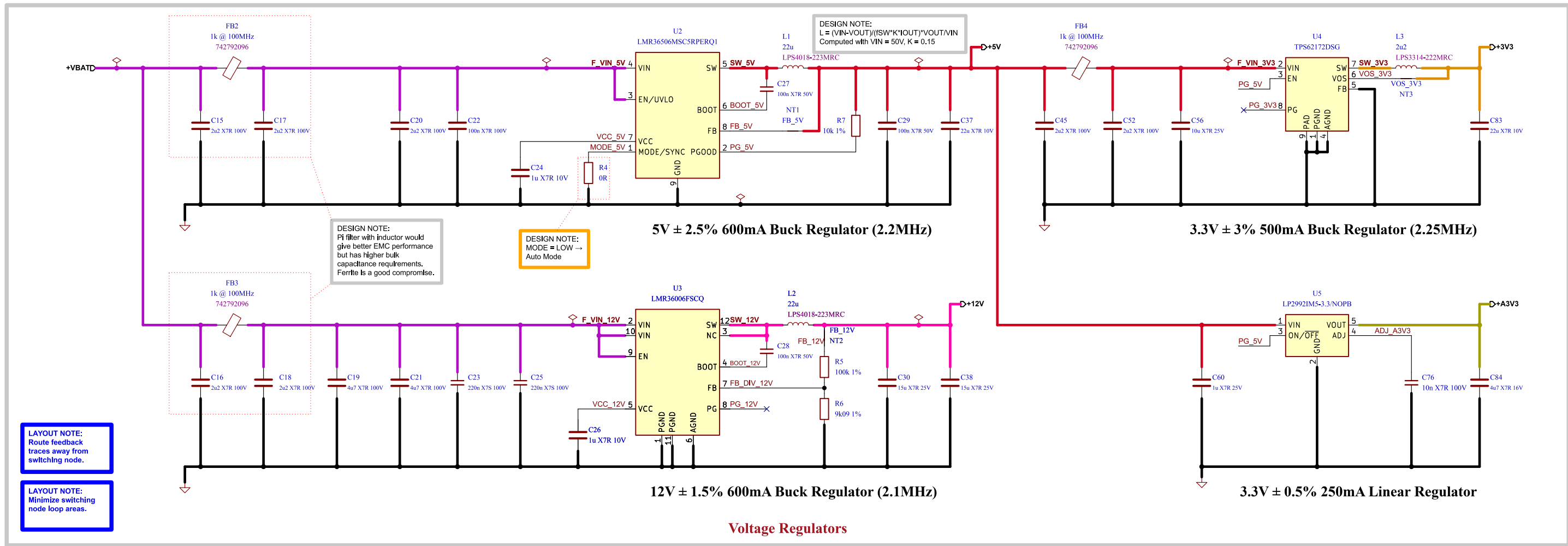
### Revision:

1.1

### Sheet:

5 of 21

# [6] Power - Generation



**Battery**

- +VBAT TP39 +VBAT
- +VBAT TP40 +VBAT
- +VBAT TP41 +VBAT

**Filtered Power Rails**

- F\_VIN\_12V TP42 F\_VIN\_12V
- F\_VIN\_5V TP43 F\_VIN\_5V
- F\_VIN\_3V3 TP44 F\_VIN\_3V3

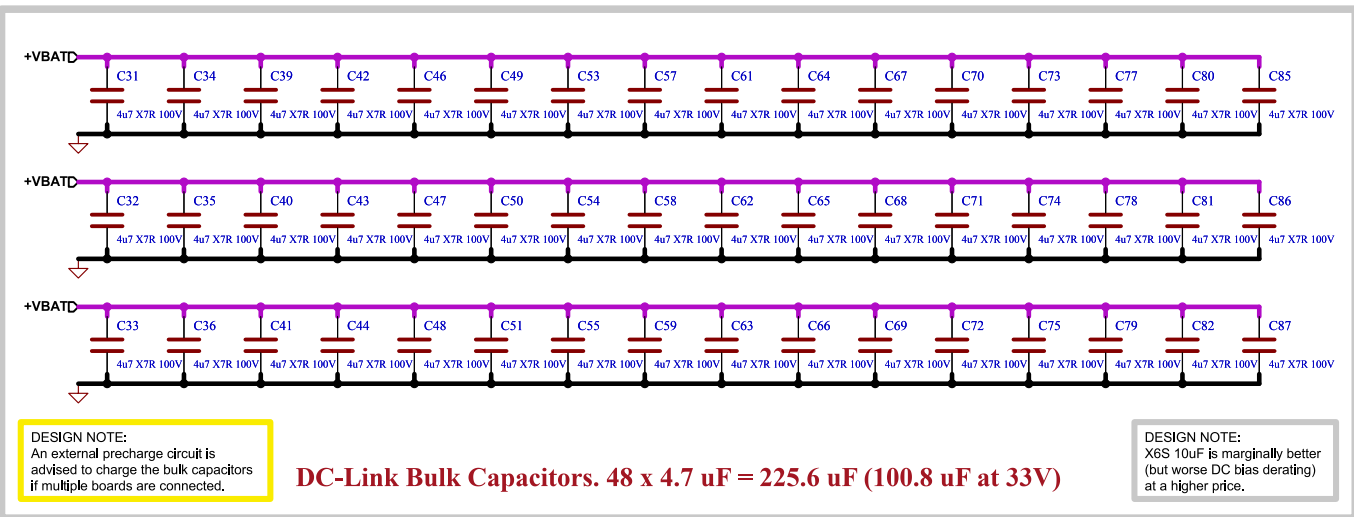
**Power Rails**

- +12V TP45 +12V
- +5V TP46 +5V
- +5V TP47 +5V
- +3V3 TP48 +3V3
- +3V3 TP49 +3V3
- +3V3 TP50 +3V3
- +A3V3 TP51 +A3V3

**Ground**

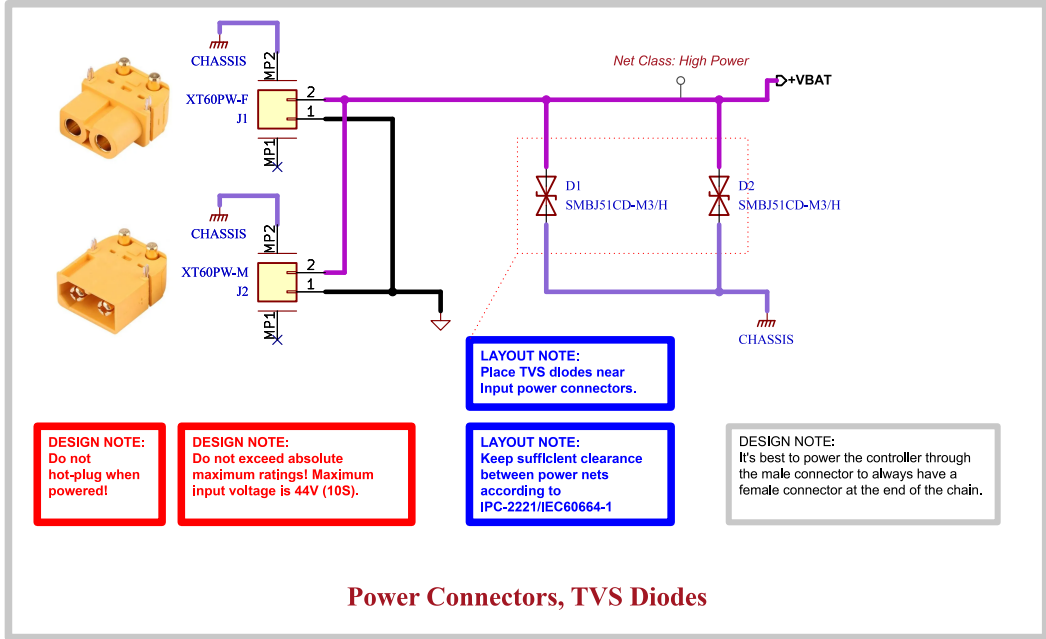
- GND TP52 GND
- GND TP53 GND
- GND TP54 GND
- GND TP55 GND
- GND TP56 GND
- GND TP57 GND
- GND TP58 GND

**Test Points**



|                                                                                                                                               |  |                                                   |                                    |                                    |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------|------------------------------------|------------------------------------|-------------------------|
| <b>Comments:</b><br>+12V (600 mA): Fan power<br>+5V (600 mA): Gate Driver, LEDs<br>+3V3 (500 mA): digital 3.3V<br>+3V3A (250 mA): analog 3.3V |  | <b>Company:</b><br>EPFL Xplore Research ABC       |                                    | <b>Variant:</b><br>NO_SNUBBER      |                         |
| <b>Sheet Title:</b><br>Power - Generation                                                                                                     |  | <b>Board Name:</b><br>Amulet Motion Controller    |                                    | <b>Project Name:</b><br>Chienpanzé |                         |
| <b>Sheet Path:</b><br>/Project Architecture/Power - Generation/                                                                               |  | <b>File Name:</b><br>Power - Generation.kicad_sch | <b>Designer:</b><br>Vincent Nguyen | <b>Date:</b><br>2024-04-13         | <b>Revision:</b><br>1.1 |
|                                                                                                                                               |  | <b>Reviewer:</b>                                  | <b>Size:</b><br>A3                 | <b>Sheet:</b><br>6 of 21           |                         |

## [7] Power - Connectors



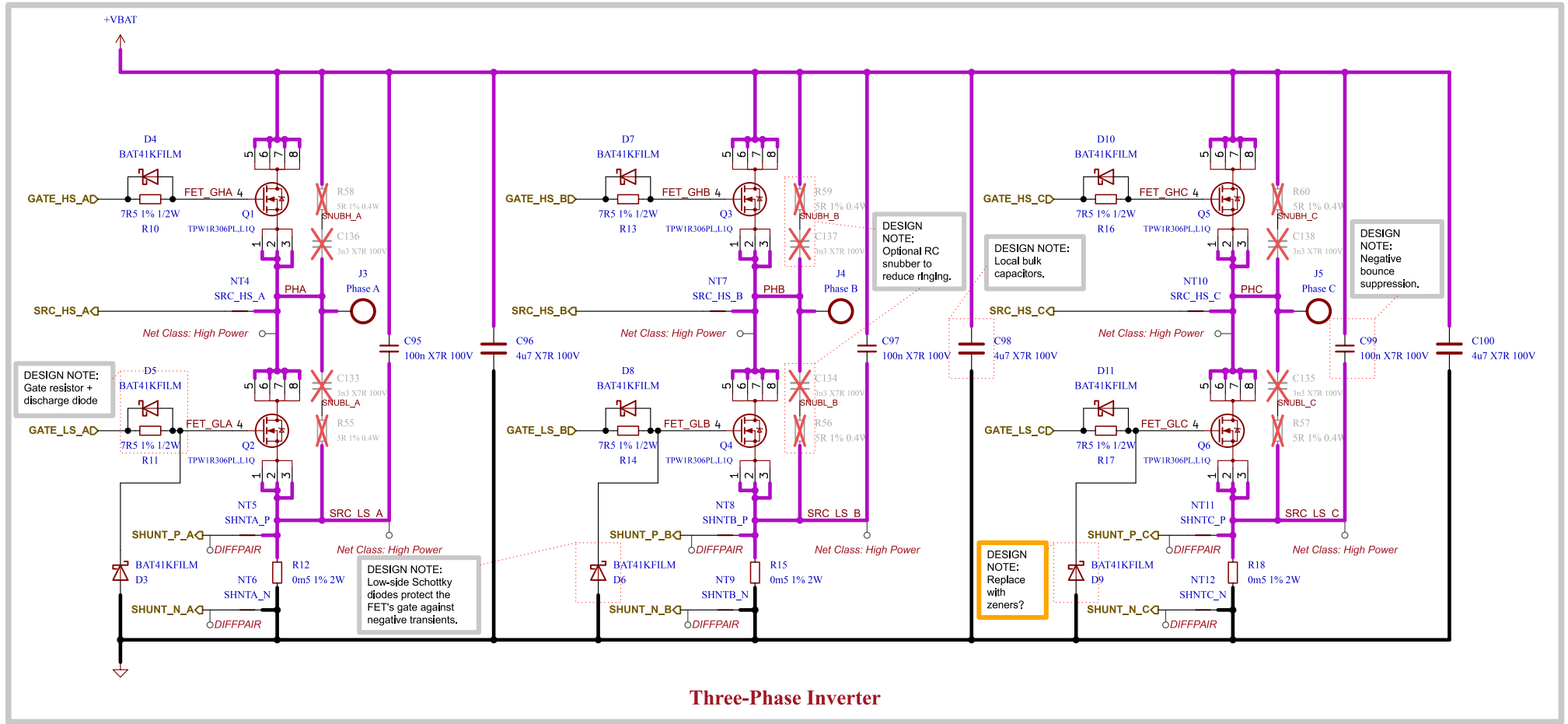
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|                                                          |                                    | Board Name:<br>Amulet Motion Controller    |                             |                                                                                       | Project Name:<br>Chienpanzé |  |
|                                                          | Sheet Title:<br>Power - Connectors | File Name:<br>Power - Connectors.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-31                                                                   | Revision:<br>1.1            |  |
| Sheet Path:<br>/Project Architecture/Power - Connectors/ |                                    |                                            | Reviewer:                   | Size:<br>A4                                                                           | Sheet:<br>7 of 21           |  |

## 100-V Three-Phase Smart Gate Driver



|                                                                 |                                           |                                                   |                             |                                                                                       |                                    |                                 |
|-----------------------------------------------------------------|-------------------------------------------|---------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|------------------------------------|---------------------------------|
|                                                                 | Comments:                                 | Company:<br>EPFL Xplore Research ABC              |                             |  | Variant:<br>NO_SNUBBER             |                                 |
|                                                                 |                                           | Board Name:<br><b>Amulet Motion Controller</b>    |                             |                                                                                       | Project Name:<br><b>Chienpanzé</b> |                                 |
|                                                                 | Sheet Title:<br>Motor Control - Top Level | File Name:<br>Motor Control - Top Level.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-20                                                                   | Revision:<br>1.1                   |                                 |
| Sheet Path:<br>/Project Architecture/Motor Control - Top Level/ |                                           |                                                   | Reviewer:                   |                                                                                       | Size:<br><b>A3</b>                 | Sheet:<br><b>8</b> of <b>21</b> |

# [9] Motor Control - Inverter



**LAYOUT NOTE:**  
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

**LAYOUT NOTE:**  
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

**DESIGN NOTE:**  
A gate drive current that is too large can damage the FETs!

**Comments:**  
System Design Considerations for High-Power Motor Driver Applications  
Best Practices for Board Layout of Motor Drivers  
Proper RC Snubber Design for Motor Drivers

**Sheet Title:**  
Motor Control - Inverter

**Sheet Path:**  
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

**Company:**  
EPFL Xplore Research ABC



**Board Name:**  
Amulet Motion Controller

**File Name:**  
Motor Control - Inverter.kicad\_sch

**Designer:**  
Vincent Nguyen

**Reviewer:**

**Variant:**  
NO\_SNUBBER

**Project Name:**  
Chienpanzé

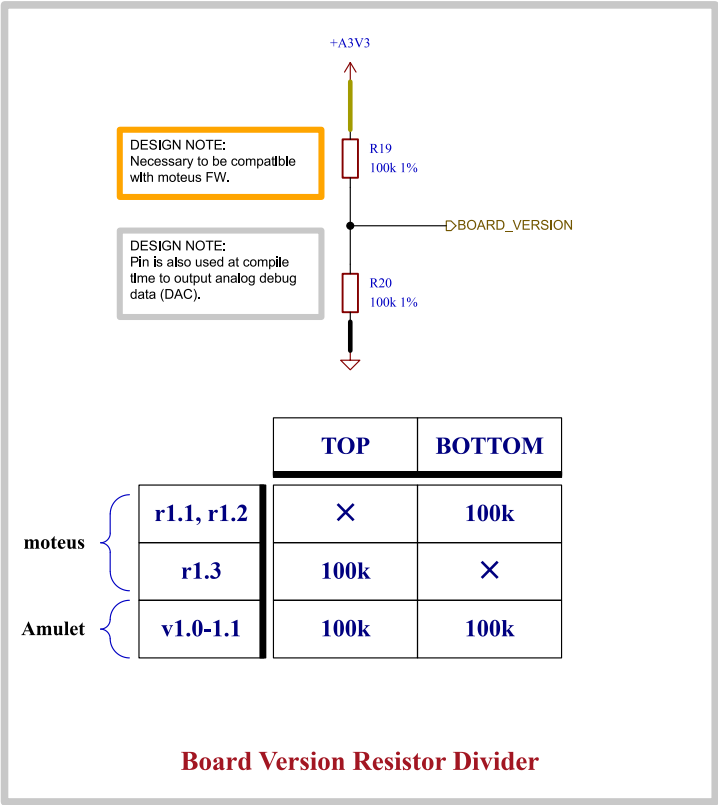
**Date:**  
2024-01-25

**Size:**  
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**Revision:**  
1.1

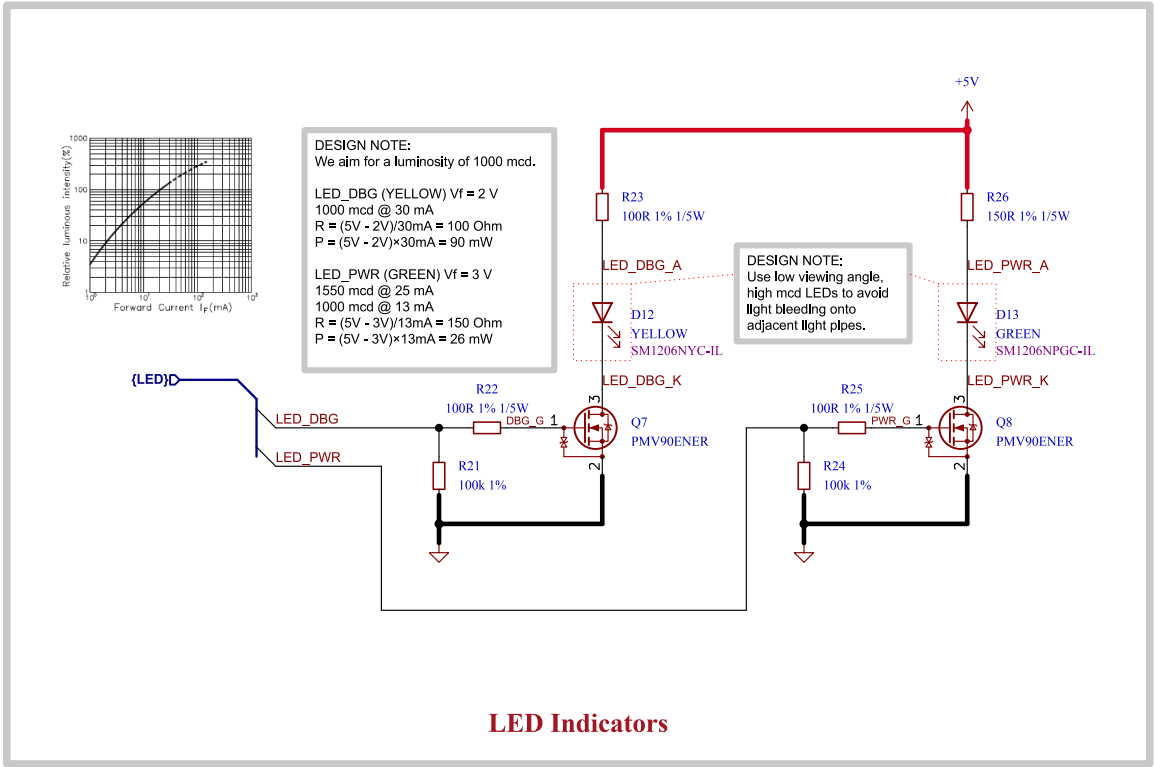
**Sheet:**  
9 of 21

# [10] Misc - Board Version, DAC



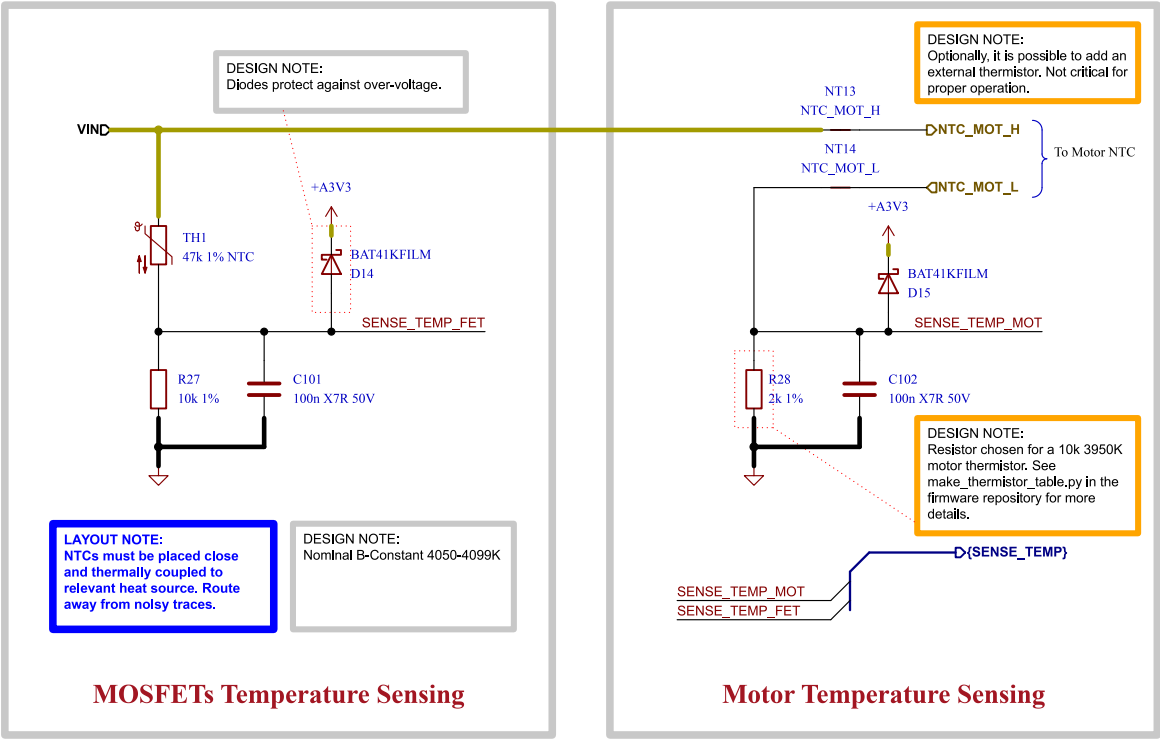
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|  |                                                              | Board Name: Amulet Motion Controller          |                          | Project Name: Chienpanzé |                 |
|  | Sheet Title: Misc - Board Version, DAC                       | File Name: Misc - Board Version DAC.kicad_sch | Designer: Vincent Nguyen | Date: 2024-04-13         | Revision: 1.1   |
|  | Sheet Path: /Project Architecture/Misc - Board Version, DAC/ |                                               | Reviewer:                | Size: A4                 | Sheet: 10 of 21 |

# [11] User - LED Indicators



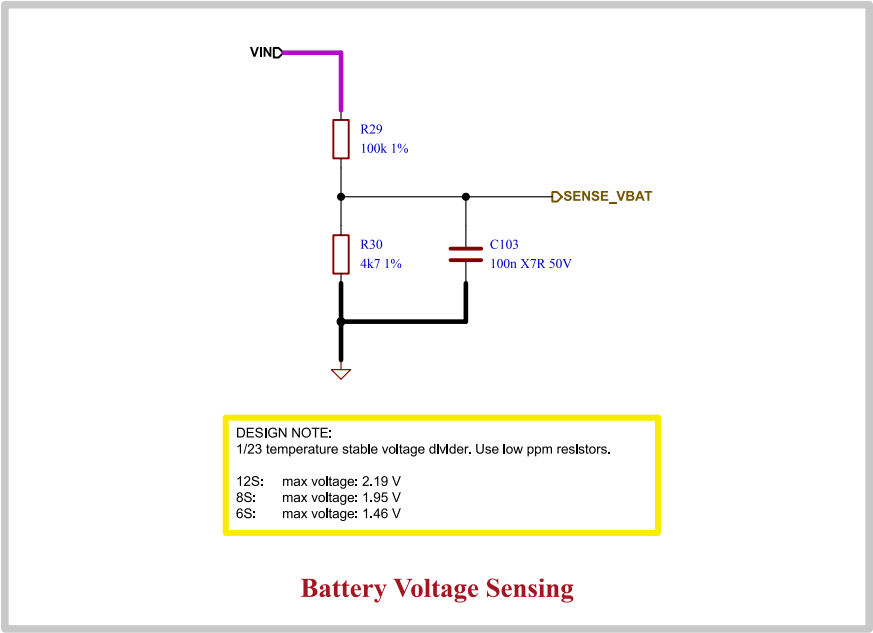
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|  | Comments:                                                   | Company:<br>EPFL Xplore Research ABC          |                             | Variant:<br>NO_SNUBBER      |                    |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             | Project Name:<br>Chienpanzé |                    |
|  | Sheet Title:<br>User - LED Indicators                       | File Name:<br>User - LED Indicators.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-19         | Revision:<br>1.1   |
|  | Sheet Path:<br>/Project Architecture/User - LED Indicators/ |                                               | Reviewer:                   | Size:<br>A4                 | Sheet:<br>11 of 21 |

[12] Sensing - Temperature



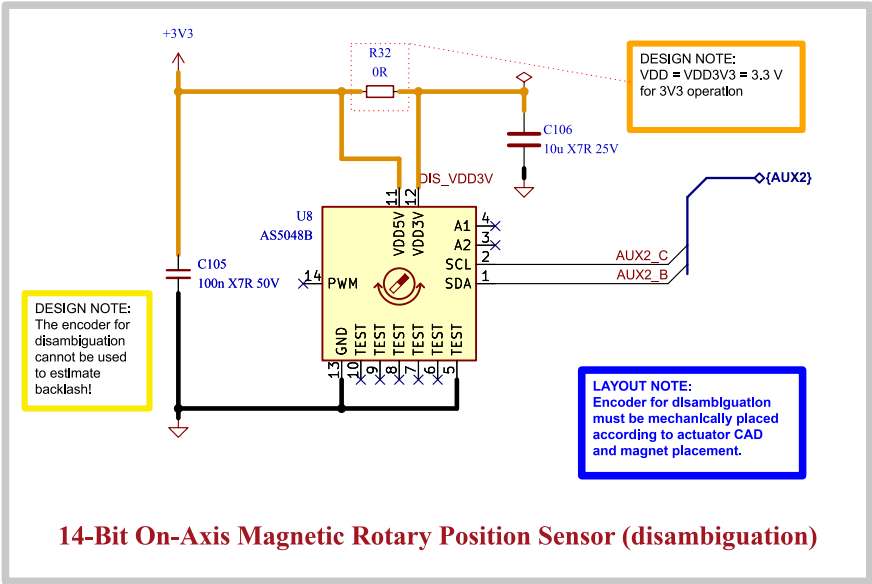
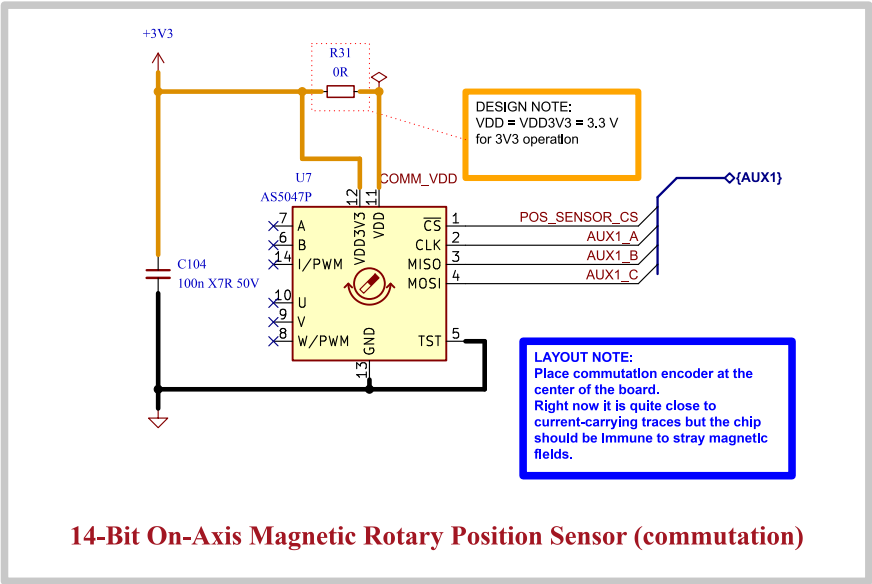
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|  | Comments:                                                   | Company:<br>EPFL Xplore Research ABC          |                             |  | Variant:<br>NO_SNUBBER      |                    |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             |  | Project Name:<br>Chienpanzé |                    |
|  | Sheet Title:<br>Sensing - Temperature                       | File Name:<br>Sensing - Temperature.kicad_sch | Designer:<br>Vincent Nguyen |  | Date:<br>2024-04-13         | Revision:<br>1.1   |
|  | Sheet Path:<br>/Project Architecture/Sensing - Temperature/ |                                               | Reviewer:                   |  | Size:<br>A4                 | Sheet:<br>12 of 21 |

# [13] Sensing - Battery



|  |                                                         |  |                                           |                             |                             |                    |
|--|---------------------------------------------------------|--|-------------------------------------------|-----------------------------|-----------------------------|--------------------|
|  | Comments:                                               |  | Company:<br>EPFL Xplore Research ABC      |                             | Variant:<br>NO_SNUBBER      |                    |
|  |                                                         |  | Board Name:<br>Amulet Motion Controller   |                             | Project Name:<br>Chienpanzé |                    |
|  | Sheet Title:<br>Sensing - Battery                       |  | File Name:<br>Sensing - Battery.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-10-14         | Revision:<br>1.1   |
|  | Sheet Path:<br>/Project Architecture/Sensing - Battery/ |  |                                           | Reviewer:                   | Size:<br>A4                 | Sheet:<br>13 of 21 |

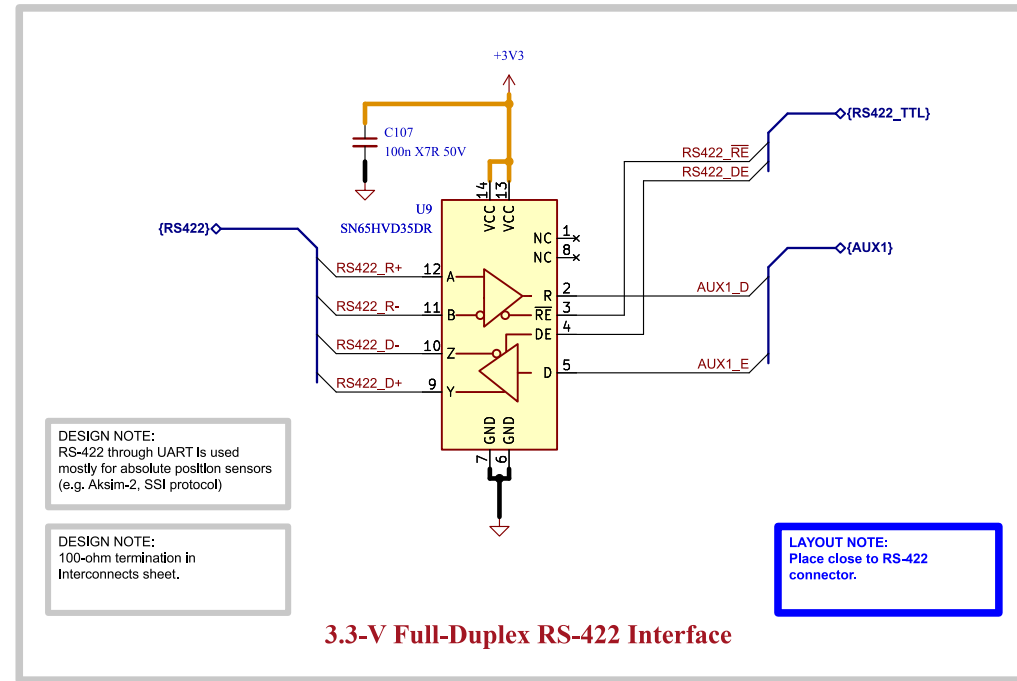
# [14] Sensing - Position



**DESIGN NOTE:**  
AS5047P senses magnet mounted on planetary sun gear, for commutation.  
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

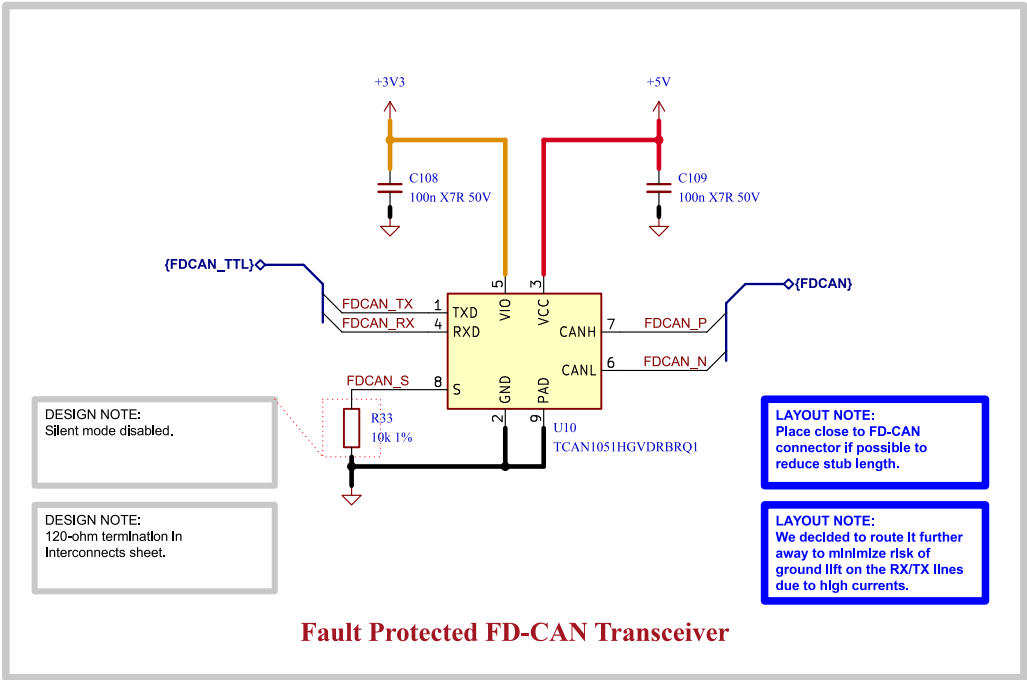
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|                                           | Comments: |                              | Company:                 |                |  |            | Variant:      |           |  |
|                                           |           |                              | EPFL Xplore Research ABC |                |                                                                                       |            | NO_SNUBBER    |           |  |
|                                           |           |                              | Board Name:              |                |                                                                                       |            | Project Name: |           |  |
|                                           |           |                              | Amulet Motion Controller |                |                                                                                       |            | Chienpanzé    |           |  |
| Sheet Title:                              |           | File Name:                   |                          | Designer:      |                                                                                       | Date:      |               | Revision: |  |
| Sensing - Position                        |           | Sensing - Position.kicad_sch |                          | Vincent Nguyen |                                                                                       | 2023-10-14 |               | 1.1       |  |
| Sheet Path:                               |           |                              |                          | Reviewer:      |                                                                                       | Size:      |               | Sheet:    |  |
| /Project Architecture/Sensing - Position/ |           |                              |                          |                |                                                                                       | A4         |               | 14 of 21  |  |
| 3                                         |           | 4                            |                          | 5              |                                                                                       |            |               | 6         |  |

# [15] Interface - RS-422



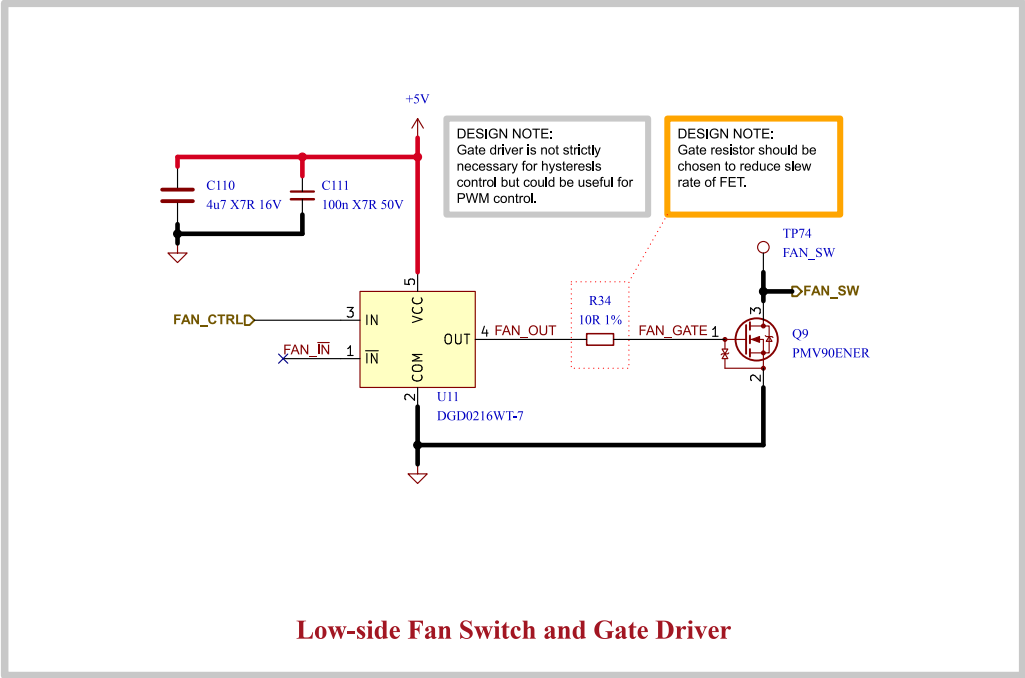
|  |                                                          |                                            |                                                |                     |                                    |  |
|--|----------------------------------------------------------|--------------------------------------------|------------------------------------------------|---------------------|------------------------------------|--|
|  | Comments:                                                |                                            | Company:<br>EPFL Xplore Research ABC           |                     | Variant:<br>NO_SNUBBER             |  |
|  |                                                          |                                            | Board Name:<br><b>Amulet Motion Controller</b> |                     | Project Name:<br><b>Chienpanzé</b> |  |
|  | Sheet Title:<br>Interface - RS-422                       | File Name:<br>Interface - RS-422.kicad_sch | Designer:<br>Vincent Nguyen                    | Date:<br>2023-10-15 | Revision:<br>1.1                   |  |
|  | Sheet Path:<br>/Project Architecture/Interface - RS-422/ |                                            | Reviewer:                                      | Size:<br><b>A4</b>  | Sheet:<br><b>15</b> of <b>21</b>   |  |

# [16] Interface - FD-CAN



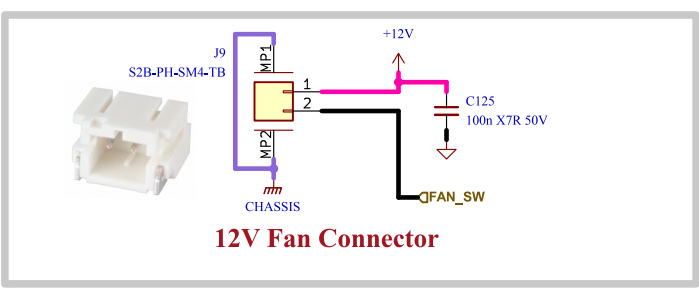
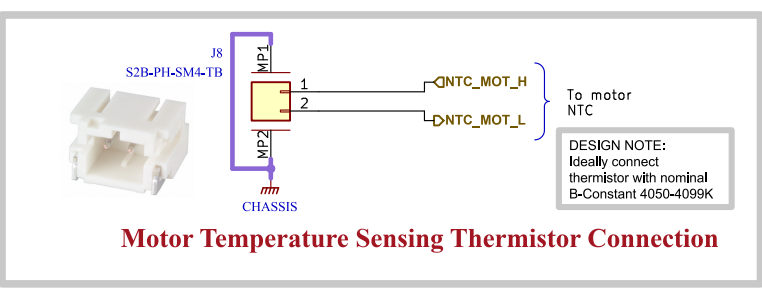
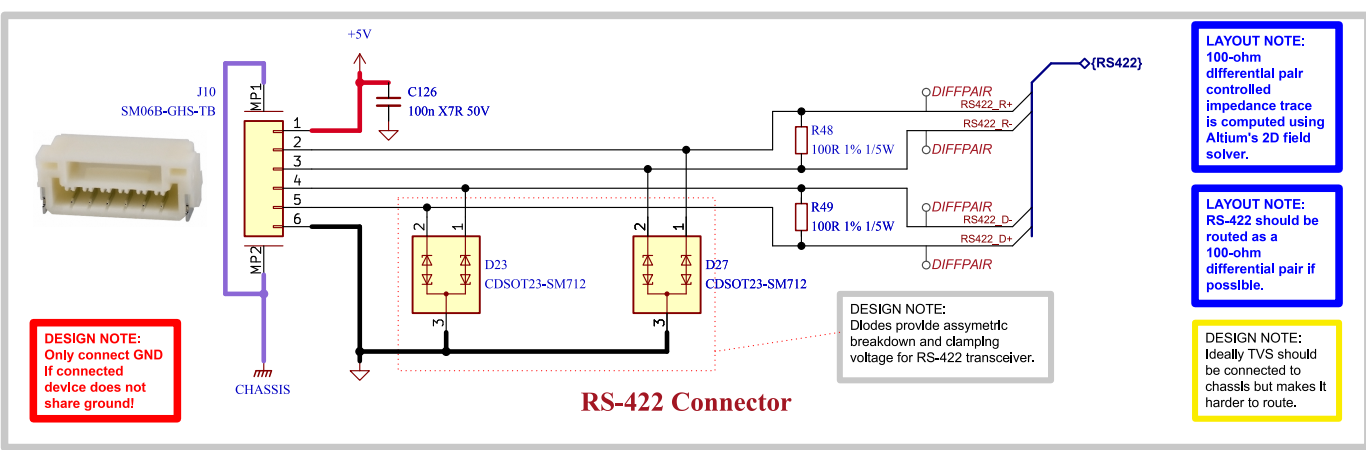
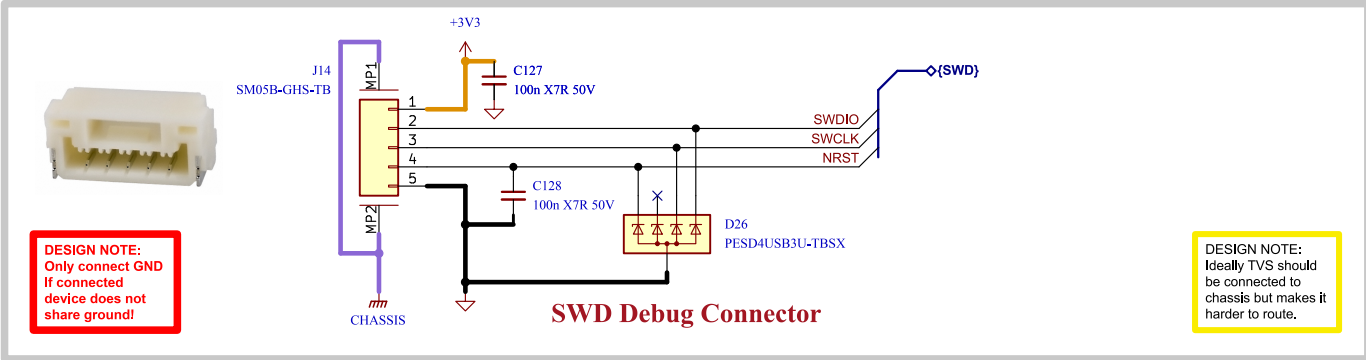
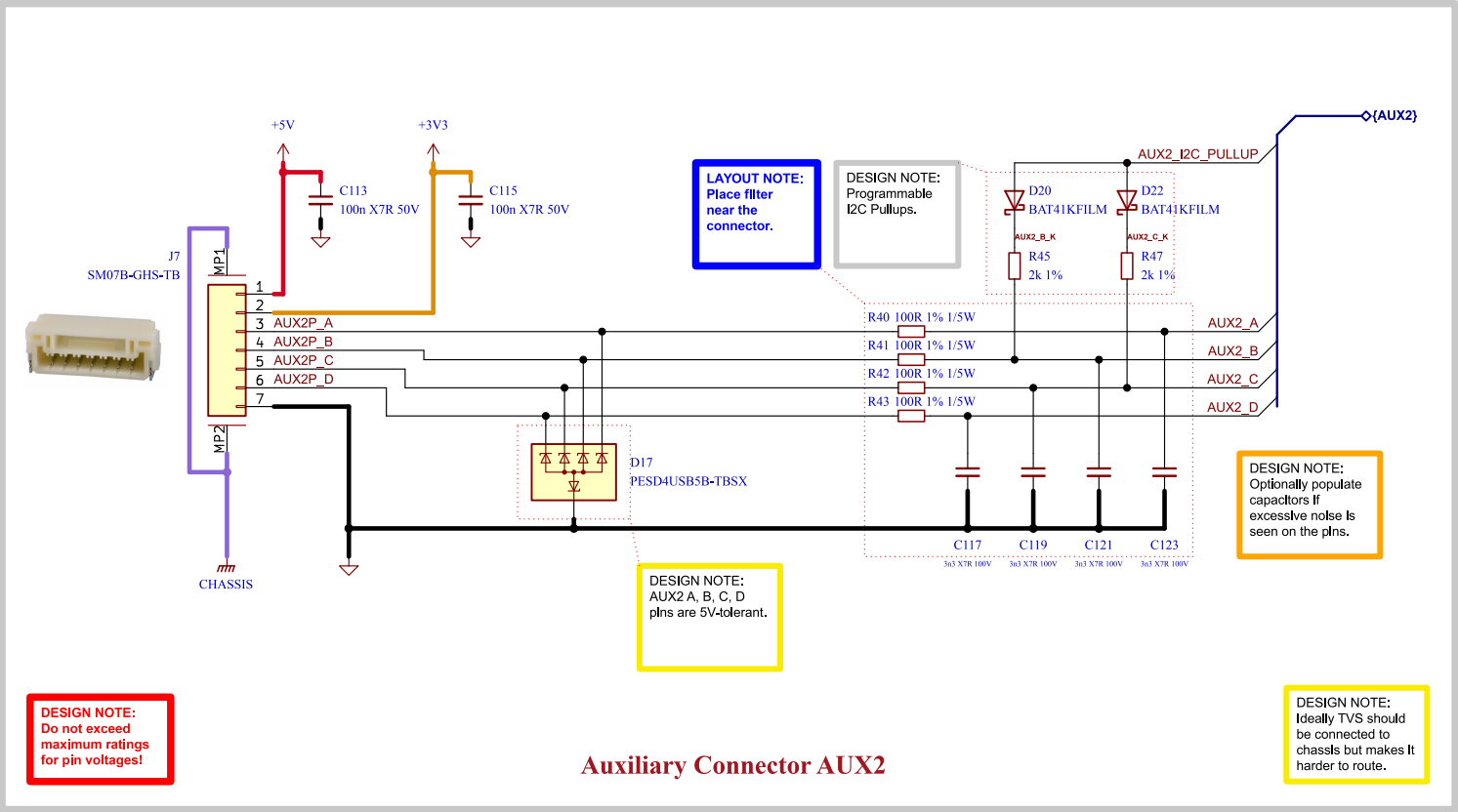
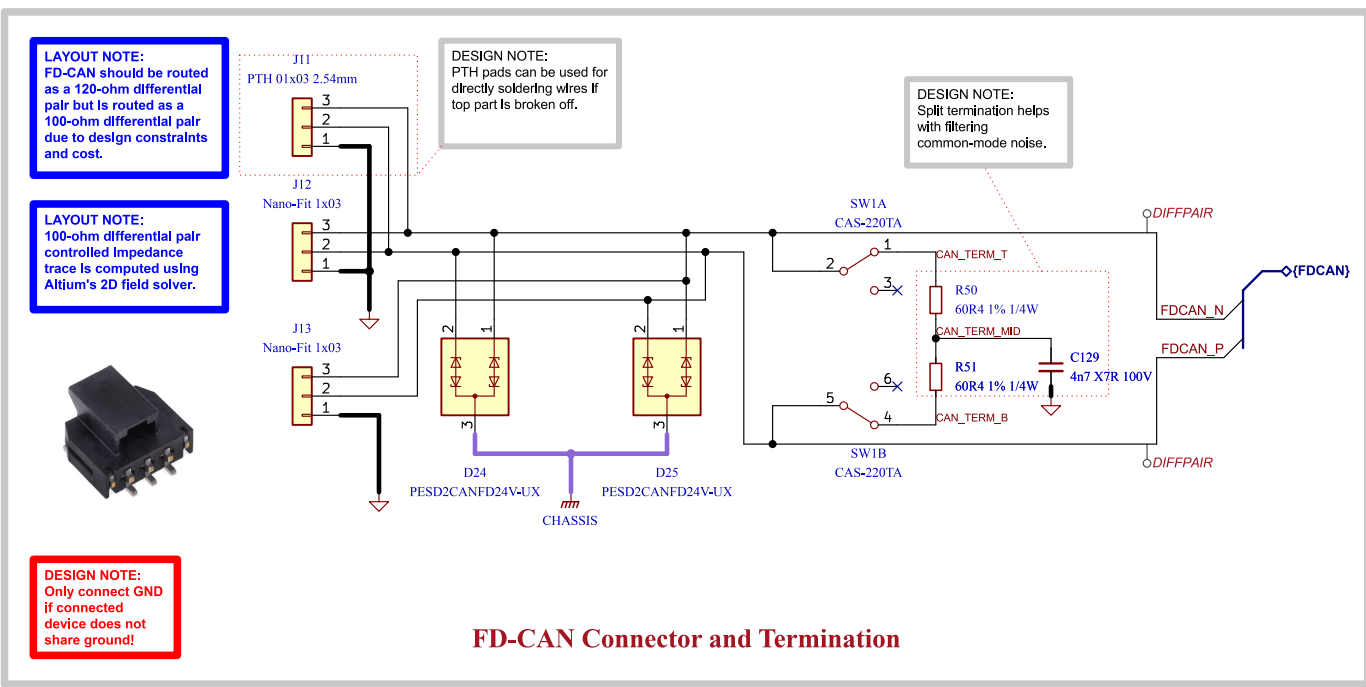
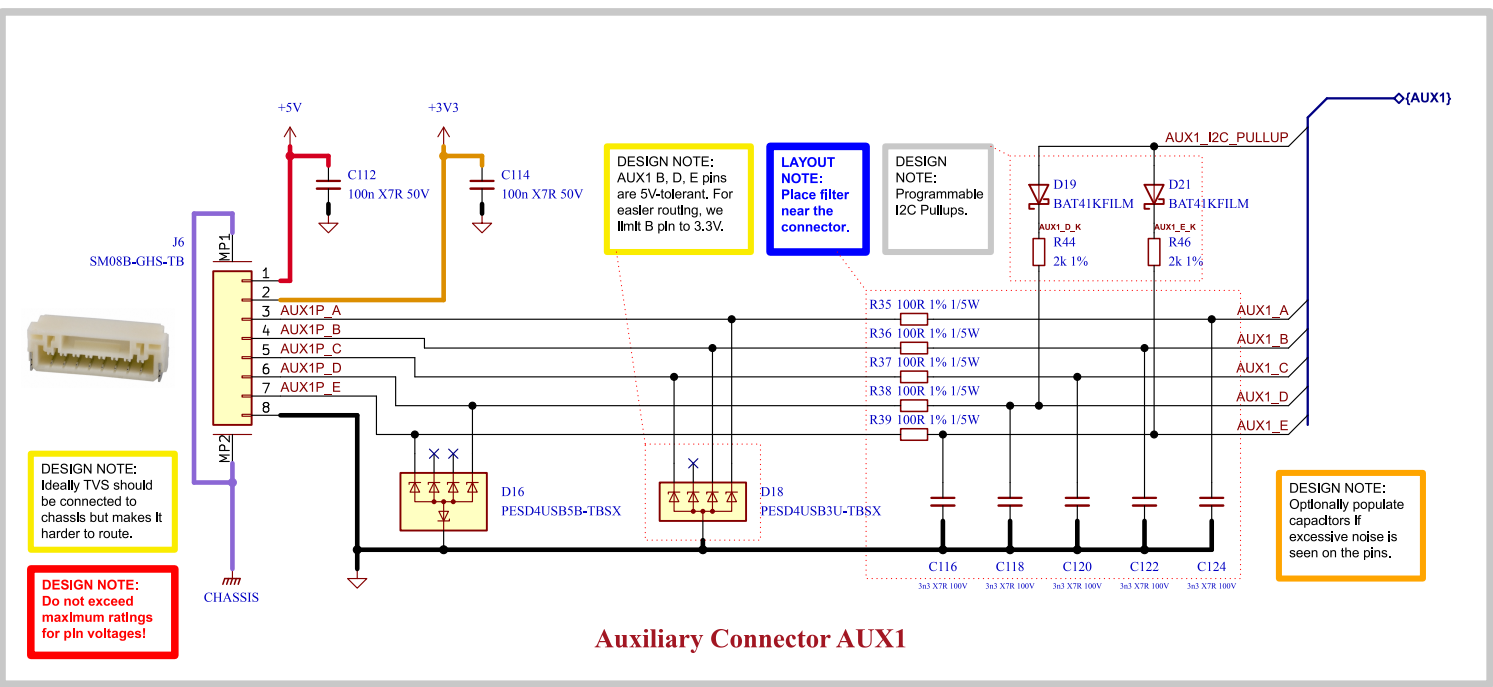
|  |                                                          |  |                                            |                             |                             |                    |
|--|----------------------------------------------------------|--|--------------------------------------------|-----------------------------|-----------------------------|--------------------|
|  | Comments:                                                |  | Company:<br>EPFL Xplore Research ABC       |                             | Variant:<br>NO_SNUBBER      |                    |
|  |                                                          |  | Board Name:<br>Amulet Motion Controller    |                             | Project Name:<br>Chienpanzé |                    |
|  | Sheet Title:<br>Interface - FD-CAN                       |  | File Name:<br>Interface - FD-CAN.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-10-15         | Revision:<br>1.1   |
|  | Sheet Path:<br>/Project Architecture/Interface - FD-CAN/ |  |                                            | Reviewer:                   | Size:<br>A4                 | Sheet:<br>16 of 21 |

# [17] Interface - Fan Control



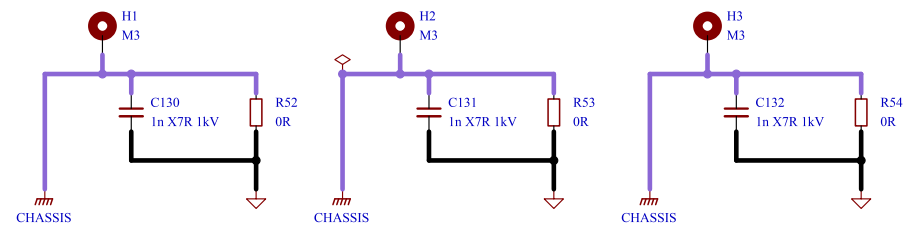
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|--|---------------------------------------------------------------|-------------------------------------------------|-----------------------------|-----------------------------|--------------------|
|  | Comments:                                                     | Company:<br>EPFL Xplore Research ABC            |                             | Variant:<br>NO_SNUBBER      |                    |
|  |                                                               | Board Name:<br>Amulet Motion Controller         |                             | Project Name:<br>Chienpanzé |                    |
|  | Sheet Title:<br>Interface - Fan Control                       | File Name:<br>Interface - Fan Control.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-11-19         | Revision:<br>1.1   |
|  | Sheet Path:<br>/Project Architecture/Interface - Fan Control/ |                                                 | Reviewer:                   | Size:<br>A4                 | Sheet:<br>17 of 21 |

# [18] Interface - Interconnects



|                                                         |                                                                 |                                                   |                             |                     |
|---------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------|-----------------------------|---------------------|
| Comments:<br>Reference:<br>Flexible I/O worked examples | Company:<br>EPFL Xplore Research ABC                            |                                                   | Variant:<br>NO_SNUBBER      |                     |
|                                                         | Board Name:<br>Amulet Motion Controller                         |                                                   | Project Name:<br>Chienpanzé |                     |
|                                                         | Sheet Title:<br>Interface - Interconnects                       | File Name:<br>Interface - Interconnects.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-31 |
|                                                         | Sheet Path:<br>/Project Architecture/Interface - Interconnects/ | Reviewer:                                         | Size:<br>A3                 | Revision:<br>1.1    |
|                                                         |                                                                 |                                                   | Sheet:<br>18 of 21          |                     |

# [19] Misc - Holes, Fiducials



DESIGN NOTE:  
Optional RC filter for chassis  
grounding. Capacitors offer multi-point  
low-impedance connections at  
high-frequencies.

DESIGN NOTE:  
0-Ohm resistor provides low-impedance  
bond between chassis and signal GND.  
Be aware of ground loops through cable  
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

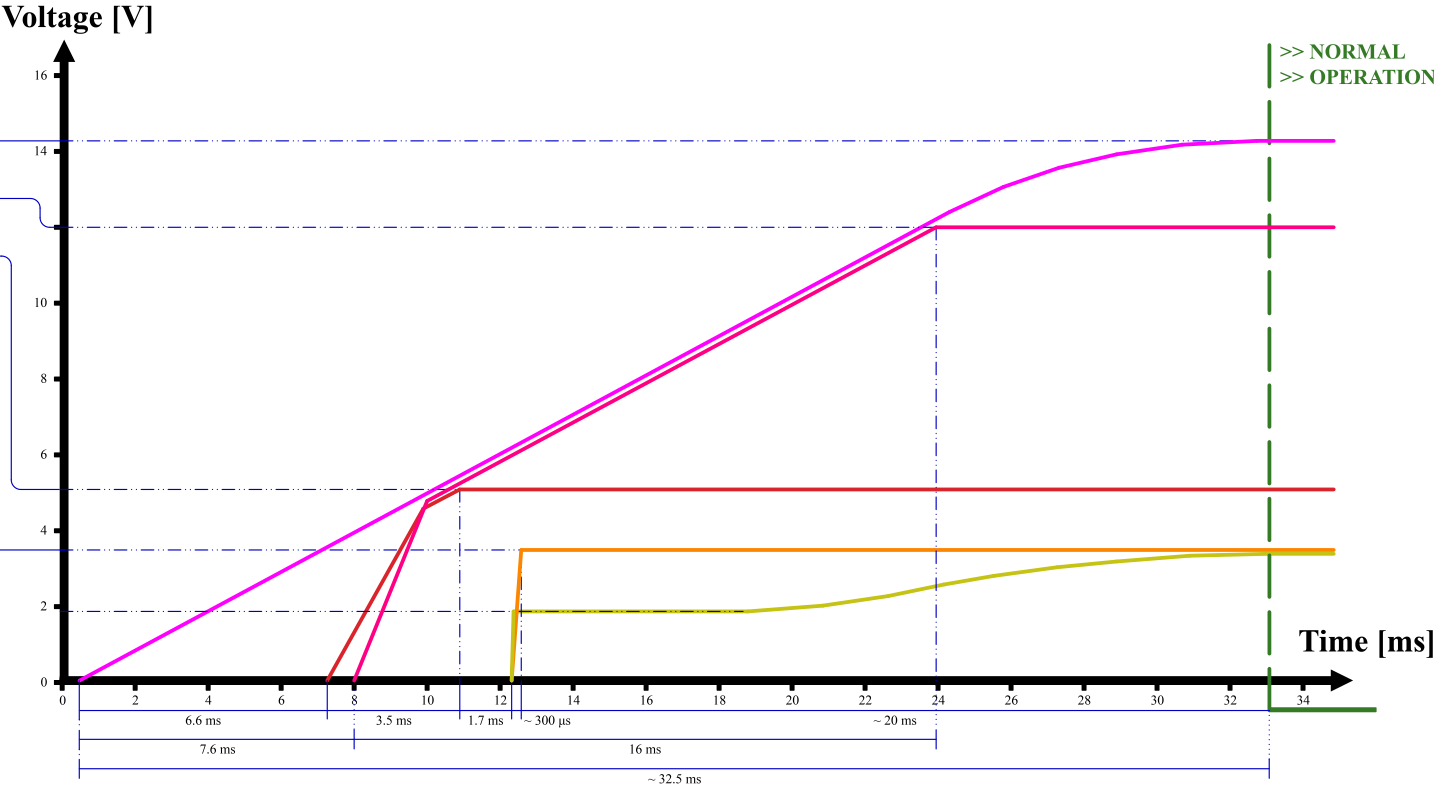
|  |                                                               |                                                |                                         |                     |                             |  |
|--|---------------------------------------------------------------|------------------------------------------------|-----------------------------------------|---------------------|-----------------------------|--|
|  | Comments:                                                     |                                                | Company:<br>EPFL Xplore Research ABC    |                     | Variant:<br>NO_SNUBBER      |  |
|  |                                                               |                                                | Board Name:<br>Amulet Motion Controller |                     | Project Name:<br>Chienpanzé |  |
|  | Sheet Title:<br>Misc - Holes, Fiducials                       | File Name:<br>Misc - Holes Fiducials.kicad_sch | Designer:<br>Vincent Nguyen             | Date:<br>2023-10-22 | Revision:<br>1.1            |  |
|  | Sheet Path:<br>/Project Architecture/Misc - Holes, Fiducials/ |                                                | Reviewer:                               | Size:<br>A4         | Sheet:<br>19 of 21          |  |

# [20] Power - Sequencing

| NAME  | SOURCE   | LEVEL       |
|-------|----------|-------------|
| +VBAT | BATTERY  | 12 - 44V    |
| +12V  | LMR36006 | 12V ± 1.5%  |
| +5V   | LMR36506 | 5V ± 1.5%   |
| +3V3  | TPS62172 | 3.3V ± 3%   |
| +A3V3 | LP2992   | 3.3V ± 0.5% |

DESIGN NOTE:  
Power sequencing graph measured experimentally.

DESIGN NOTE:  
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



|  |                                     |                                            |                             |                     |
|--|-------------------------------------|--------------------------------------------|-----------------------------|---------------------|
|  | Comments:                           | Company:<br>EPFL Xplore Research ABC       | Variant:<br>NO_SNUBBER      |                     |
|  |                                     | Board Name:<br>Amulet Motion Controller    | Project Name:<br>Chienpanzé |                     |
|  | Sheet Title:<br>Power - Sequencing  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-03-12 |
|  | Sheet Path:<br>/Power - Sequencing/ | Reviewer:                                  | Size:<br>A4                 | Revision:<br>1.1    |
|  |                                     |                                            | Sheet:<br>20 of 21          |                     |

## [21] Revision History

**12-DEC-2023 - Initial Release**  
**Variant: v1.0 Preliminary**

- Changed CPH-CPL capacitor to 47nF (gate driver).
- Changed FD-CAN transceiver IC.
- Changed FETs for top cooled variant.
- Added TVS protection and termination switch to FD-CAN.
- Added low-side switched 12V 600mA source for external fan.
- Added LDO for analog supply.
- Changed input power TVS diode to bidirectional and added one diode per connector.
- Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines.
- Added second onboard I2C magnetic encoder for disambiguation.
- Switched PWM\_PHASEA with PWM\_PHASEC on STM32G474 pinout for easier routing.
- Changed RS422 pinout on connector.
- Added ESD protection to all interfaces.
- Added overvoltage protection on thermistor ADC inputs.
- Changed buck regulators to optimize for low noise.
- Added Pi filters to inputs of buck regulators and MCU analog supply.
- Added decoupling caps next to power pins of connectors.

**25-JAN-2024 - First Revision**  
**Variant: v1.0 Checked**

- Added controller target specifications.
- Replaced 5V 300mA buck converter with 600mA version.
- Added credits to moteus on cover page.
- Added optional RC-Snubber to power stage.
- Increased chassis length to go around the board.
- CAN and power TVS diodes now go to chassis.
- Changed clearance between nets to respect IEC60664-1 where possible.
- Rectified comment on precharge.
- Changed power TVS diode reference designator from "U" to "D".
- Replaced chassis-GND capacitor by 1nF 1kV.

**12-MAR-2024 - First Revision**  
**Variant: v1.0 Released**

- Modified power sequencing graph according to experimental data.

**13-APR-2024 - Second Revision**  
**Variant: v1.1 Released**

- Added RC snubber passive values.
- Added more vias for VBUS and LMR36006 GND pads.
- Changed board version voltage reference from +3V3 to +A3V3.
- Changed motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

|                                   |                                          |                             |                                                                                       |                             |                        |
|-----------------------------------|------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|------------------------|
| Comments:                         | Company:<br>EPFL Xplore Research ABC     |                             |  |                             | Variant:<br>NO_SNUMBER |
|                                   | Board Name:<br>Amulet Motion Controller  |                             |                                                                                       | Project Name:<br>Chienpanzé |                        |
| Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-01-03                                                                   | Revision:<br>1.1            |                        |
| Sheet Path:<br>/Revision History/ |                                          | Reviewer:                   | Size:<br>A4                                                                           | Sheet:<br>21 of 21          |                        |